



NAME:

SECTION: HU ID:

LINEAR ALGEBRA

SPRING 2026

QUIZ 10

Max Marks: 10

Time: 10 minutes

Q. Let S be a finite set of vectors in a finite-dimensional vector space V . If S spans V but is not a basis for V , explain in your own words how S can be converted into a basis for V . Then, prove your statement. Make sure to include the details of any theorem(s) as part of your working/justification.



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SOLUTION QUIZ 10

THEOREM 5.4.6

Let S be a finite set of vectors in a finite-dimensional vector space V .

- (a) If S spans V but is not a basis for V , then S can be reduced to a basis for V by removing appropriate vectors from S .
- (b) If S is a linearly independent set that is not already a basis for V , then S can be enlarged to a basis for V by inserting appropriate vectors into S .

Proof (a) If S is a set of vectors that spans V but is not a basis for V , then S is a linearly dependent set. Thus some vector $\mathbf{v}$ in S is expressible as a linear combination of the other vectors in S . By the Plus/Minus Theorem (Theorem 5.4.4b), we can remove $\mathbf{v}$ from S , and the resulting set S' will still span V . If S' is linearly independent, then S' is a basis for V , and we are done. If S' is linearly dependent, then we can remove some appropriate vector from S' to produce a set S'' that still spans V . We can continue removing vectors in this way until we finally arrive at a set of vectors in S that is linearly independent and spans V . This subset of S is a basis for V .

Proof (b) Suppose that $\dim(V) = n$. If S is a linearly independent set that is not already a basis for V , then S fails to span V , and there is some vector $\mathbf{v}$ in V that is not in $\text{span}(S)$. By the Plus/Minus Theorem (Theorem 5.4.4a), we can insert $\mathbf{v}$ into S , and the resulting set S' will still be linearly independent. If S' spans V , then S' is a basis for V , and we are finished. If S' does not span V , then we can insert an appropriate vector into S' to produce a set S'' that is still linearly independent. We can continue inserting vectors in this way until we reach a set with n linearly independent vectors in V . This set will be a basis for V by Theorem 5.4.5.

THEOREM 5.4.4

Plus/Minus Theorem

Let S be a nonempty set of vectors in a vector space V .

- (a) If S is a linearly independent set, and if $\mathbf{v}$ is a vector in V that is outside of $\text{span}(S)$, then the set $S \cup \{\mathbf{v}\}$ that results by inserting $\mathbf{v}$ into S is still linearly independent.
- (b) If $\mathbf{v}$ is a vector in S that is expressible as a linear combination of other vectors in S , and if $S - \{\mathbf{v}\}$ denotes the set obtained by removing $\mathbf{v}$ from S , then S and $S - \{\mathbf{v}\}$ span the same space; that is,

$$\text{span}(S) = \text{span}(S - \{\mathbf{v}\})$$

MARKING SCHEME



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[3] marks for fully correct statement in 5.4.6 part a and b for version I and II respectively

[3] marks for stating correct use of the plus/minus theorem

[3] marks for mentioning repeating the process till S becomes a basis

[1] mark for fully correct work, good notation etc. Deducted if 1 or 2 minor slips in reasoning, or poor notation seen.